

SCORING, SUBSTITUTION MATRICES AND SCORE AGGREGATION FOR TCR ANTIGEN-SPECIFICITY SEARCH

The `vdjmatch` authors*

vdjmatch technical appendix — draft, regenerated from `bench/gen_vdjam.py`, `bench/loo_vdjam.py`,
`bench/gen_appendix_assets.py`

Abstract

`vdjmatch` annotates T-cell receptor (TCR) clonotypes by antigen specificity through fuzzy CDR3 search against a labelled database (VDJdb), built on the `seqtree` search core. This note documents the *scoring* layer: how CDR3 similarity is scored, the data-driven TCR substitution matrix (VDJAM) and why it is region-structured, and how per-hit scores are aggregated into a control-calibrated significance. CDR3 loops resist the assumptions behind BLOSUM/PAM — they cannot be multiply aligned across clonotypes of different length, and germline-encoded flanks (CASS...) are so over-conserved that a naive matrix conflates germline bias with selection. We give a position-debiased log-odds estimator that isolates the substitution signal, show empirically that this signal lives in the non-template (NDN) core while the V/J flanks are germline-determined, and demonstrate the consequences on worked examples (GIL, NLV) using the alignments and CIGAR strings produced by `seqtree`. The E-value theory used for significance is derived in the companion `seqtree` appendix; here we summarize it and give the paired-chain extension.

1. THE SEARCH-AND-SCORE FRAMEWORK

Given a query CDR3 q and a labelled reference set D (VDJdb), `vdjmatch` retrieves the neighbours of q within a fixed scope $\theta = (s, i, d, t)$ (max substitutions, insertions, deletions, total edits) using the `seqtree` engine, which defines a ball $B_\theta(q) = \{x : \text{edit}(q, x) \leq \theta\}$ and returns, per hit, the edit decomposition and a matrix-weighted alignment. The matrix enters as a *ranking* score inside the ball; the ball itself (an edit budget) bounds the candidate set. Each hit carries its antigen labels, an extended CIGAR (`=/X/I/D`), and an alignment score. Significance comes from comparing the neighbour count to a matched background control (§3).

2. SUBSTITUTION SCORING AND THE VDJAM MATRIX

WHY BLOSUM/PAM DO NOT TRANSFER.. BLOSUM is a log-odds matrix estimated from gapless columns of a trusted multiple alignment of homologous blocks. CDR3s violate both premises: (i) same-specificity CDR3s differ in length and have no column homology, so there is no multiple

*Supplementary technical note for the `vdjmatch` software; intended as supplementary material to the main manuscript.

alignment; (ii) the germline-encoded ends are over-conserved — essentially every human TRB CDR3 begins CASS and ends in a short J motif (...F), so an alanine near the N-terminus is germline-fixed and carries no specificity information, while an alanine in the centre does. A single global matrix estimated over all positions conflates germline conservation with selection.

OBSERVATION UNIT AND ESTIMATOR.. We replace “alignment columns” with *same-antigen single-substitution pairs*: ordered pairs of CDR3s annotated to the same epitope, of equal length, differing at exactly one position — the only clean substitution signal extractable without a multiple alignment. For a substitution $a \leftrightarrow b$ observed at position p in a CDR3 of length L on chain c , let $\text{bg}(\cdot \mid L, p)$ be the residue frequency at position p among all length- L chain- c CDR3s. With $f(a, b)$ the observed count of $\{a, b\}$ events and

$$e(a, b) = \sum_{\text{event slots } (L, p)} n_{L, p} [\text{bg}(a \mid L, p) \text{bg}(b \mid L, p) + \text{bg}(b \mid L, p) \text{bg}(a \mid L, p)], \quad \text{VDJAM}(a, b) = \log_2 \frac{f(a, b) + \alpha}{e(a, b) + \alpha}, \quad (1)$$

where $n_{L, p}$ is the number of observed events at (L, p) and α a Laplace pseudocount. The key point is that the background is *position-specific*: at an over-conserved column the residue frequency is peaked, so substitutions there are both rare (small f) and “expected” (matched e), and contribute ≈ 0 to the log-odds — the germline bias is removed analytically rather than by hand-masking. The matrix is consumed by `seqtree` via the squared-distance penalty $\text{pen}(a, b) = s_{aa} + s_{bb} - 2s_{ab}$ (`SubstitutionMatrix.from_similarity`); a dominant uniform diagonal makes higher learned similarity rank a substitution better.

REGION STRUCTURE: NDN IS LEARNED, V/J ARE GERMLINE-DETERMINED.. A CDR3 is V-germline prefix · NDN insert · J-germline suffix. Estimating (1) separately by region shows (Fig. 1) that the substitution signal — the agreement with general amino-acid chemistry, measured as correlation with BLOSUM62 — is concentrated in the NDN core (TRB $r=0.42$, TRA 0.35), whereas the V flank carries almost none (TRB 0.08, TRA 0.07): V-flank “substitutions” are dominated by germline allele differences, not free chemistry. This motivates a two-part model: **VDJAM is learned for the NDN core**, while the **V and J flanks are auto-computed** from the germline V/J sequences and their trimming. Concretely, a CDR3 position derived from V- (or J-) germline is near-invariant when it survives trimming and reverts to the random-insert background when trimmed; so the V/J contribution is a per-position conservation profile $w(p) = \text{Pr}[\text{position } p \text{ is germline-retained}]$, determined by the gene’s trimming-length distribution, rather than a substitution matrix learned from data. Combined, the NDN matrix scores the core and $w(p)$ down-weights substitutions at germline-retained flanks (realized through a per-position weighting of the base matrix). *Status*: the NDN matrix and the region decomposition are implemented and evaluated below; the germline+trimming V/J profile (from the OLGA recombination model, via `mirpy`) is in progress.

3. SCORE AGGREGATION AND SIGNIFICANCE

PER-HIT TO PER-QUERY.. A region-aware similarity score sums the NDN matrix score over the core and the (down-weighted) flank contributions; this is available as an optional ranking score and generalizes the legacy cloglog generalized-linear aggregation over V/CDR1/CDR2/J/CDR3.

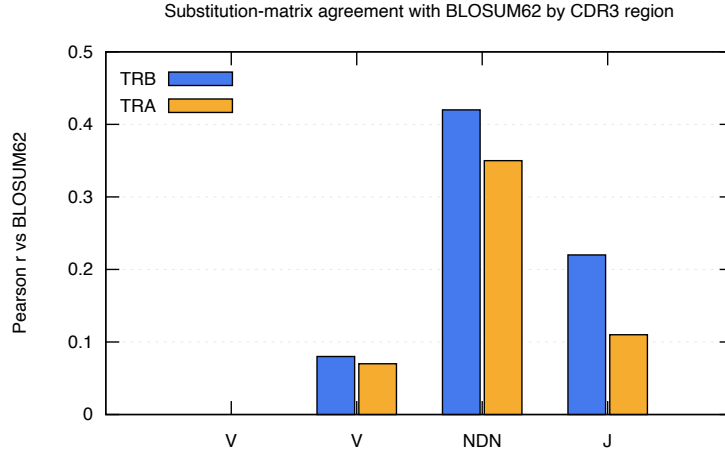


Figure 1: Per-region agreement of the learned CDR3 substitution matrix with BLOSUM62 (Pearson r of off-diagonal scores), human VDJdb. The free-substitution signal lives in the NDN core; the V flank is germline-determined.

CONTROL-CALIBRATED E-VALUE (PRIMARY).. Immune repertoires are biologically redundant, so an i.i.d. null over-calls. `vdjmatch` instead counts a query’s VDJdb neighbours within the ball and compares to the count expected from a matched background control: with $N = |D|$, M the control size and n_C the control neighbour count, $\mathbb{E} = (N/M) n_C$ is the expected number of VDJdb neighbours by chance and $p_{\text{enrich}} = \mathbb{P}(\text{Poisson}(\mathbb{E}) \geq n_D)$ is the enrichment significance — significant only when a clonotype has *more* VDJdb neighbours than the generative background predicts. The Poisson/Chen–Stein theory, the punctured (self-excluding) null and the choice of M are derived in the companion `seqtree` appendix.

PAIRED $\alpha\beta$ E-VALUE.. For a paired query the chains are independent under the null, so the joint ball mass factorizes, $\pi_0^{\alpha\beta} \approx \pi_0^\alpha \pi_0^\beta$, and among the N paired VDJdb complexes the expected number of chance joint matches is $\mathbb{E}_{\alpha\beta} = N \pi_0^\alpha \pi_0^\beta$ with p the Poisson tail on the count of complexes matched in *both* chains. The joint null is far smaller than either chain’s, so true pairs are dramatically more significant: on the paired TCRvdb benchmark, 296/300 pairs match a complex and 295/296 (99.7%) recover the correct epitope, with joint p down to 3.6×10^{-9} .

4. RESULTS: DOES VDJAM HELP?

A matrix derived from same-antigen pairs must be tested on epitopes it never saw, or the evaluation is circular. For each held-out epitope E^* we re-derive the NDN matrix from *all other* epitopes, then run VDJdb-vs-VDJdb retrieval on E^* (exact self excluded), ranking neighbours by each matrix, and score the micro precision–recall AUC for “neighbour shares E^* ”. Across the eight largest TRB epitopes (scope 2, Fig. 2) the learned NDN matrix improves over unit (edit-distance) cost (mean $\Delta\text{PR-AUC} = +0.018$) and is on par with BLOSUM62 (-0.009 ; better on 4/8). At this scope the search ball already supplies most of the specific-vs-nonspecific discrimination and the matrix only

re-ranks within it; the larger gains are expected from the region-aware combination (NDN matrix + germline/trimming flank weighting) and at wider scopes, which is the current line of work.

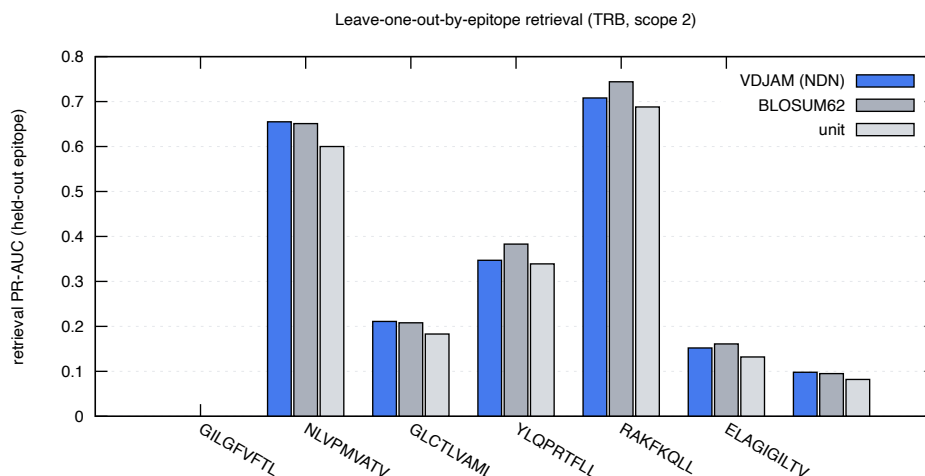


Figure 2: Leave-one-out-by-epitope retrieval PR-AUC (TRB, scope 2): the NDN VDJAM matrix versus BLOSUM62 and unit cost on epitopes excluded from matrix estimation.

5. WORKED EXAMPLES

PAIRWISE (DIFFERENT LENGTHS).. seqtree aligns CDR3s of differing length, placing indels where the loop genuinely lengthens. The edit-distance alignments below (influenza GIL-specific TRB CDR3s) show the conserved CASS V-anchor and DTQYF J-motif with substitutions and indels confined to the central NDN core:

query (ref): CASSTRATETQYF epitope GILGEVFTL (unit / edit-distance alignment)

vs CASSTRAGETQYF (len 13; subs=1 ins=0 dels=0; CIGAR 7=1X5=)

```
CASSTRATETQYF
||||||| |||||
CASSTRAGETQYF
```

vs CASSRGATQETQYF (len 14; subs=2 ins=1 dels=0; CIGAR 4=2X2=1I5=)

```
CASSTRAT-ETQYF
||||  || |||||
CASSRGATQETQYF
```

vs CASSTGQETQYF (len 12; subs=2 ins=0 dels=1; CIGAR 5=1D2X5=)

```
CASSTRATETQYF
|||||  |||||
CASST-GQETQYF
```

vs CASSYAETQYF (len 11; subs=1 ins=0 dels=2; CIGAR 4=1D1X1=1D5=)

```

CASSTRATETQYF
|||| | ||||
CASS-YA-ETQYF

```

```

vs CASSQERAGQETQYF (len 15; subs=2 ins=2 dels=0; CIGAR 4=1I1X2=1I1X5=)
CASS-TRA-TETQYF
|||| || ||||
CASSQERAGQETQYF

```

MULTIPLE (SAME LENGTH).. Aligning same-length same-epitope CDR₃s (no indels needed) makes the conservation profile explicit (Figs. 3–4): columns are shaded by conservation, dark at the germline-determined CASS prefix and J motif, light through the variable NDN core. This is the structural counterpart of the finding of Shcherbinin et al. [1] that specificity-preserving CDR₃ substitutions occur predominantly outside the peptide/MHC-contacting positions while preserving the loop backbone — i.e. the tolerated variation we score in the NDN core is exactly the variation that does not break recognition.

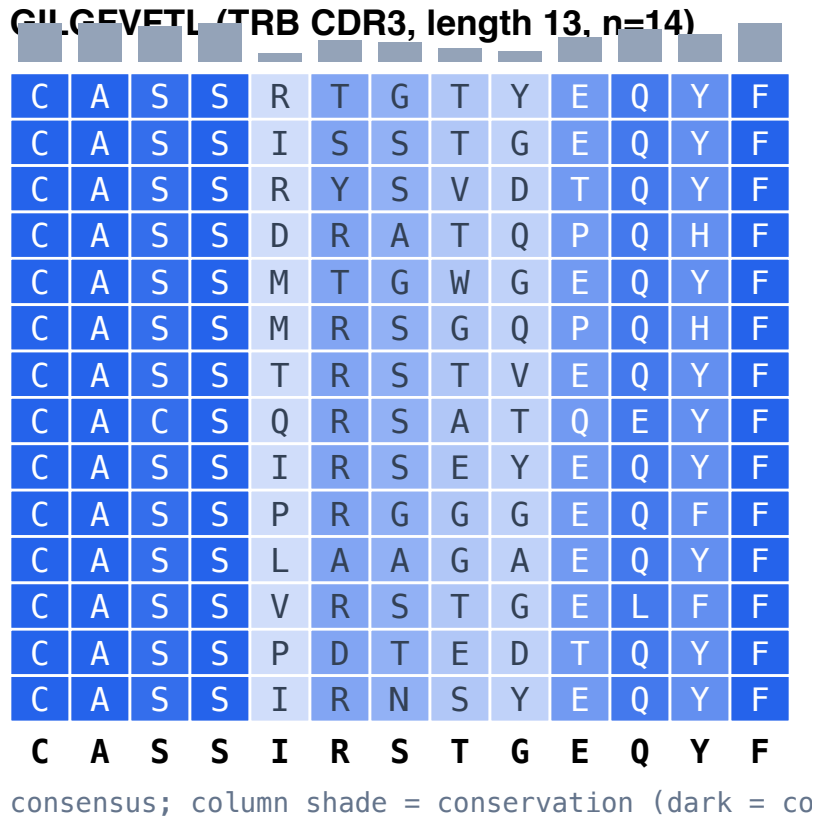


Figure 3: GIL (influenza M1, A*02) TRB CDR₃ block; conserved V/J flanks, variable NDN core.

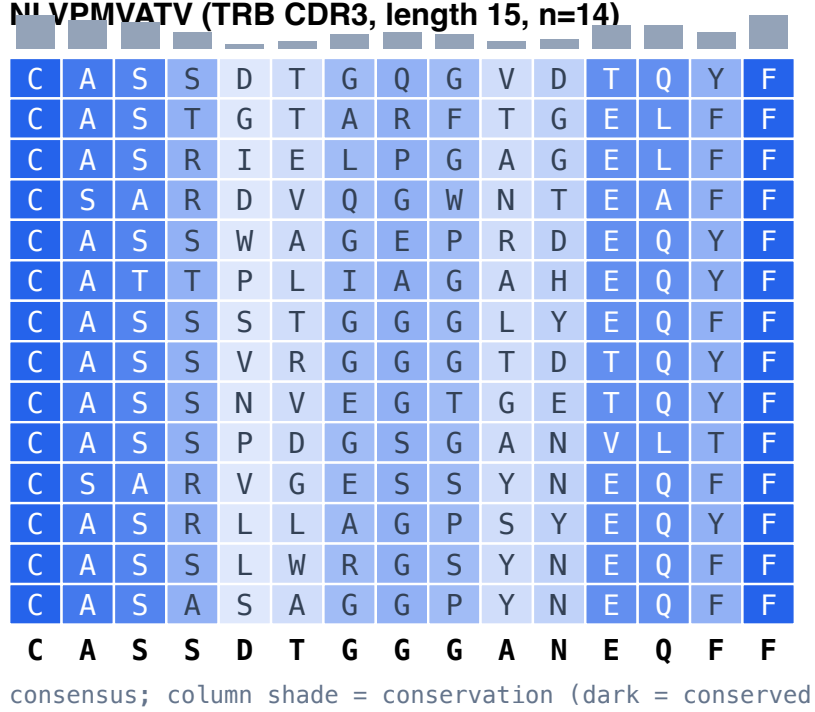


Figure 4: NLV (CMV pp65, A*02) TRB CDR3 block; same conservation pattern.

6. LIMITATIONS AND ROADMAP

The NDN matrix is estimated from a limited set of same-antigen single-substitution pairs and, at narrow scope, matches rather than beats BLOSUM62; the region-aware V/J germline+trimming model (in progress) and per-position weighting are expected to add discrimination, as is a per-chain treatment (TRA vs TRB) and a V-gene similarity term (grouping V genes by CDR1/CDR2/CDR2.5 distance). The control's V-J usage must match the query repertoire's, or the null is mis-specified; whether to renormalize V-J usage (at the risk of erasing genuine V-gene bias) is an open question evaluated on held-out epitopes. These extensions, and the full benchmark against existing tools, are the subject of ongoing work.

REFERENCES

- [1] Dmitrii S. Sheherbinin, Vadim K. Karnaukhov, Ivan V. Zvyagin, Dmitriy M. Chudakov, and Mikhail Shugay. Large-scale template-based structural modeling of T-cell receptors with known antigen specificity reveals complementarity features. *Frontiers in Immunology*, 14:1224969, 2023.